

GEOMETRY-RESOLVED BOUNDS FOR QUANTUM ESTIMATION WITH AN APPLICATION TO MAGNETIC NAVIGATION

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ABSTRACT

Magnetic navigation using quantum sensors can yield spatially-separated posterior modes when the anomaly map contains repeated structure. In this regime, fixed-tolerance success characterize global localization performance more directly than within-mode variance. With quantum Bayesian decisions, we develop geometric coarse-graining of continuous fixed-tolerance estimation. A measurable partition of the latent parameter or trajectory space define exact prior-weighted operators. Two metric hypergraphs encode the sets of cells that intersect each tolerance ball and those contained entirely within in. The corresponding finite semidefinite programs provide upper and lower bounds on the optimal radius- r success probability. The geometric approximation is given by a shift in tolerance radius no greater than the maximum cell diameter measured in the target space. The bounds tighten monotonically under refinement and converge at continuity points whenever partitions of vanishing target diameter exist. The construction lifts directly from output-state measurements to quantum channel testers. It also retains the full family of geometrically realizable lists and cannot be weaker than an unconstrained fixed-cardinality list reduction. In a stylized magnetic navigation problem with repeated map texture, receiver-matched calculations show that a locally efficient Cramér–Rao bound can be orders of magnitude too optimistic, while the geometry-resolved bound closely tracks wrong-region probability and improves on both binary Ziv–Zakai and list- M -ary comparators. An unrestricted one-interrogation calculation shows the improvement persists for a quantum converse.

1. INTRODUCTION

Migratory birds use a magnetic compass thought to involve radical-pair quantum spin dynamics [1] and can extract positional information from geomagnetic cues [2]. In engineered systems, magnetic-anomaly navigation (MagNav) estimates position by matching measured magnetic field with a pre-characterized spatial map [3]. Quantum magnetometers offer a path to improved sensitivities, and their prospective use in navigation satellite-denied operation [4] is a broader strategic interest in deployable quantum sensing [5].

The estimation problem in MagNav, however, is often not dominated by sensor noise alone. Instead, magnetic textures repeated in different regions can produce several far away map matches. A local Cramér–Rao bound (CRB) may then be small even when an estimator is likely to lock on to the wrong region; Bayesian CRBs have been used to characterize local magnetic localization accuracy [6]. Here we aim for a global bound that no admissible estimator can beat, including before the correct local posterior basin has been identified.

Classical estimation theory develops global bounds by reducing estimation to decision: Ziv and Zakai related estimation-error tails to binary testing [7], Bell et al. extended the construction to

vectors [8], and recent work develops measure-theoretic and M -ary forms [9]. List hypothesis testing permits several labels to be declared simultaneously acceptable [10]; more general finite Bayesian decisions can be represented by state-action hypergraphs [11]. For fixed-tolerance estimation, they suggests crediting an estimate for every latent cell compatible with the same tolerance ball, rather than requiring identification of a single cell.

In quantum settings, quantum statistical decision theory gives machinery to develop hypothesis-testing based estimation bounds. Holevo formulated general quantum decisions using operator-valued measures [12], and Yuen et al. characterized multiple quantum testing with arbitrary costs [13]; subset discrimination is a related restricted case [14]. Quantum Ziv–Zakai bounds replace classical binary tests by Helstrom discrimination, with related confidence-region and wave-form results [15–17]. Meyer et al. formulate finite-sample quantum metrology directly through fixed-accuracy success and a continuous semi-infinite program, together with finite discretization guarantees under regularity of the state family [18].

Here we develop bounds for fixed-tolerance quantum estimation. This builds on Bavaresco et al., who optimize single-shot Bayesian estimation by representing probe preparation and measurement as a quantum tester and solving semidefinite programs (SDP) to optimize the protocol [19]. Their framework allows general rewards, including discontinuous fixed-width confidence rewards; their explicit score-convergence argument for such rewards uses regularity of the prior.

Here we also reduce estimation to a finite quantum decision problem over multiple hypotheses and draw on the corresponding cost-matrix SDP. Our core contribution is a geometry-resolved inner-outer construction:

1. exact prior-weighted cell operators avoid point-state approximation;
2. possible and certain metric lists yield finite upper/lower bounds;
3. approximation error is bounded by tolerance-radius shift, instead of additive score errors;
4. the bounds tighten monotonically under refinement and converge at continuity points without prior-density or state-continuity assumptions; and
5. the same proof applies to quantum testers, while the full hypergraph can strictly outperform binary and fixed-cardinality list reductions.

We also demonstrate the validity of our approach in the presence of the global-alias phenomenon in a stylized MagNav setting. Comparison to baseline bounding techniques shows the advantages of our new approach.

2. FIXED-TOLERANCE QUANTUM ESTIMATION

Let $(\Theta, \mathcal{F}, \mu)$ be a probability space, let $\beta : \Theta \rightarrow (\mathbf{X}, d)$ be a measurable target functional, and let ρ_θ be a measurable density

operator on a finite-dimensional Hilbert space $\mathcal{H}$. The latent variable θ may be a scalar parameter, a complete trajectory, or a trajectory together with nuisance variables. A Borel POVM M on $\mathbf{X}$ reports an estimate x . Its radius- r Bayes success is

$$\eta_M(r) = \int_{\Theta} \text{Tr}[\rho_{\theta} M(B_r(\beta(\theta)))] \mu(d\theta),$$

$$\eta^*(r) = \sup_M \eta_M(r), \quad (1)$$

where $B_r(z) = \{x : d(x, z) \leq r\}$. Define the reward operator,

$$A_r(x) = \int_{\Theta} \mathbf{1}\{d(x, \beta(\theta)) \leq r\} \rho_{\theta} \mu(d\theta). \quad (2)$$

Tonelli's theorem gives

$$\eta_M(r) = \int_{\mathbf{X}} \text{Tr}[A_r(x) M(dx)], \quad (3)$$

which is the continuous fixed-tolerance quantum estimation problem in finite-sample quantum metrology [18].

Choose a finite measurable partition $\mathcal{C} = \{C_1, \dots, C_K\}$ of Θ . Assume

$$D_k = \overline{\beta(C_k)} \neq \emptyset \text{ is compact, } \Delta_C = \max_k \text{diam}(D_k), \quad (4)$$

and define the subnormalized composite operators

$$\Gamma_k = \int_{C_k} \rho_{\theta} \mu(d\theta), \quad \Gamma(S) = \sum_{k \in S} \Gamma_k. \quad (5)$$

For a proposed estimate x , define the possible (outer) and certain (inner) lists

$$S_r^{\exists}(x) = \{k : \inf_{z \in D_k} d(x, z) \leq r\}, \quad (6)$$

$$S_r^{\forall}(x) = \{k : \sup_{z \in D_k} d(x, z) \leq r\}. \quad (7)$$

They define finite hypergraphs

$$\mathcal{H}_r^{\diamond} = \{S_r^{\diamond}(x) : x \in \mathbf{X}\}, \quad \diamond \in \{\exists, \forall\}. \quad (8)$$

For a finite hypergraph $\mathcal{H} \subseteq 2^{[K]}$, define its optimal quantum decision value over POVMs $\{Q_S\}$ as,

$$V_{\Gamma}(\mathcal{H}) = \max_{\substack{Q_S \succeq 0 \\ \sum_{S \in \mathcal{H}} Q_S = I}} \sum_{S \in \mathcal{H}} \text{Tr}[Q_S \Gamma(S)]. \quad (9)$$

Set

$$\underline{\eta}_C(r) = V_{\Gamma}(\mathcal{H}_r^{\forall}), \quad \bar{\eta}_C(r) = V_{\Gamma}(\mathcal{H}_r^{\exists}). \quad (10)$$

The first value rewards only cells wholly contained in a tolerance ball; the second rewards every cell that touches it.

Lemma 1 (List enlargement). *Let $A, B : \mathbf{X} \rightarrow 2^{[K]}$ satisfy $A(x) \subseteq B(x)$ for every x . Then*

$$V_{\Gamma}(\{A(x) : x \in \mathbf{X}\}) \leq V_{\Gamma}(\{B(x) : x \in \mathbf{X}\}). \quad (11)$$

Proof. For every realized A -list S , choose x_S with $A(x_S) = S$ and put $T(S) = B(x_S)$. Given a feasible list POVM $\{Q_S\}$, define $\tilde{Q}_T = \sum_{S: T(S)=T} Q_S$. This is a feasible POVM on realized B -lists. Since $S \subseteq T(S)$, $\Gamma(S) \preceq \Gamma(T(S))$, relabeling cannot decrease the objective. Maximizing proves (11). $\square$

3. GEOMETRY-RESOLVED APPROXIMATION

Now we consider geometry-resolved approximation.

Theorem 1 (Certified metric-hypergraph approximation). *For every $r \geq 0$,*

$$\underline{\eta}_C(r) \leq \eta^*(r) \leq \bar{\eta}_C(r) \leq \underline{\eta}_C(r + \Delta_C) \leq \eta^*(r + \Delta_C). \quad (12)$$

For $r \geq \Delta_C$,

$$\eta^*(r - \Delta_C) \leq \bar{\eta}_C(r - \Delta_C) \leq \underline{\eta}_C(r). \quad (13)$$

Both $\underline{\eta}_C(r)$ and $\bar{\eta}_C(r)$ are nondecreasing in r . Moreover,

$$V_{\Gamma}(\mathcal{H}) = \min_{Y=Y^{\dagger}} \{\text{Tr } Y : Y \succeq \Gamma(S), S \in \mathcal{H}\} \quad (14)$$

for $\mathcal{H} = \mathcal{H}_r^{\exists}$ or $\mathcal{H}_r^{\forall}$, with strong duality. If $\mathcal{C}'$ is a finite measurable refinement of $\mathcal{C}$, then

$$\underline{\eta}_C(r) \leq \underline{\eta}_{C'}(r) \leq \eta^*(r) \leq \bar{\eta}_{C'}(r) \leq \bar{\eta}_C(r). \quad (15)$$

Consequently, if $\mathcal{C}_1 \preceq \mathcal{C}_2 \preceq \dots$ are nested refinements with $\Delta_{C_n} \rightarrow 0$, then at every continuity point $r > 0$ of the monotone function η^ ,*

$$\underline{\eta}_{C_n}(r) \uparrow \eta^*(r), \quad \bar{\eta}_{C_n}(r) \downarrow \eta^*(r). \quad (16)$$

Proof. Because D_k is compact, the maps $x \mapsto \inf_{z \in D_k} d(x, z)$ and $x \mapsto \sup_{z \in D_k} d(x, z)$ are continuous. Hence every list-outcome region is Borel.

Fix a POVM M in (1) and group its outcomes by their outer lists:

$$Q_S = M(\{x : S_r^{\exists}(x) = S\}), \quad S \in \mathcal{H}_r^{\exists}. \quad (17)$$

If $\theta \in C_k$ and $d(x, \beta(\theta)) \leq r$, then $\beta(\theta) \in D_k$ and $k \in S_r^{\exists}(x)$. Therefore

$$\begin{aligned} \eta_M(r) &\leq \sum_{S \in \mathcal{H}_r^{\exists}} \sum_{k \in S} \int_{C_k} \text{Tr}(\rho_{\theta} Q_S) \mu(d\theta) \\ &= \sum_{S \in \mathcal{H}_r^{\exists}} \text{Tr}[Q_S \Gamma(S)] \leq \bar{\eta}_C(r). \end{aligned} \quad (18)$$

Taking the supremum over M proves $\eta^*(r) \leq \bar{\eta}_C(r)$.

Conversely, let $\{Q_S : S \in \mathcal{H}_r^{\forall}\}$ be feasible in (9). For each realized list choose x_S such that $S_r^{\forall}(x_S) = S$, and construct the atomic estimator POVM

$$M_Q(A) = \sum_{S: x_S \in A} Q_S. \quad (19)$$

If $\theta \in C_k$ and $k \in S$, then every point of D_k , in particular $\beta(\theta)$, lies within radius r of x_S . Thus

$$\begin{aligned} \eta_{M_Q}(r) &\geq \sum_{S \in \mathcal{H}_r^{\forall}} \sum_{k \in S} \int_{C_k} \text{Tr}(\rho_{\theta} Q_S) \mu(d\theta) \\ &= \sum_{S \in \mathcal{H}_r^{\forall}} \text{Tr}[Q_S \Gamma(S)]. \end{aligned} \quad (20)$$

Maximizing over $\{Q_S\}$ gives $\underline{\eta}_C(r) \leq \eta^*(r)$.

For any nonempty compact D ,

$$\sup_{z \in D} d(x, z) \leq \inf_{z \in D} d(x, z) + \text{diam}(D). \quad (21)$$

Indeed, choose a minimizer $z_0 \in D$ and use $d(x, z) \leq d(x, z_0) + d(z_0, z)$. Hence

$$S_r^\exists(x) \subseteq S_{r+\Delta_C}^\forall(x), \quad (22)$$

and, for $r \geq \Delta_C$,

$$S_{r-\Delta_C}^\exists(x) \subseteq S_r^\forall(x). \quad (23)$$

Lemma 1, followed by the middle bounds just proved, yields (12)–(13). The same lemma applied to the pointwise list inclusions at radii $r_1 \leq r_2$ proves monotonicity in r .

For the SDP dual, introduce a Hermitian multiplier Y for $\sum_S Q_S = I$. The Lagrangian is

$$\mathcal{L}(\{Q_S\}, Y) = \text{Tr } Y + \sum_{S \in \mathcal{H}} \text{Tr}[(\Gamma(S) - Y)Q_S]. \quad (24)$$

Its supremum over $Q_S \succeq 0$ is finite exactly when $Y \succeq \Gamma(S)$ for every S , giving (14). The point $Q_S = I/|\mathcal{H}|$ is strictly feasible, so Slater's condition gives strong duality.

Finally let $\mathcal{C}' = \{C'_j\}$ refine $\mathcal{C}$, with parent map $g(j) = k$ when $C'_j \subseteq C_k$. Since the refinement is finite,

$$\Gamma_k = \sum_{j:g(j)=k} \Gamma'_j, \quad D_k = \bigcup_{j:g(j)=k} D'_j. \quad (25)$$

For outer lists, $g(S_{r,C'}^\exists(x)) = S_{r,C}^\exists(x)$. Relabeling a refined-list POVM by its coarse parent enlarges each reward operator, so $\bar{\eta}_{C'}(r) \leq \bar{\eta}_C(r)$. For inner lists, every child of a coarse-certain cell is refined-certain. Relabeling a coarse-list POVM by the refined list realized at the same representative point enlarges reward, so $\underline{\eta}_{C'}(r) \geq \underline{\eta}_C(r)$. This proves (15). Combining refinement monotonicity with (12)–(13) and using continuity of η^* at r proves (16) by squeezing. $\square$

The theorem gives a horizontal accuracy certificate. It does not assert convergence at a jump of $r \mapsto \eta^*(r)$, and it presupposes that the target image admits finite partitions whose maximum diameter tends to zero. These qualifications may be relevant for discrete and mixed priors and for non-totally-bounded target spaces.

For a finite report set $G \subset \mathbf{X}$, define

$$V_G^\diamond = \max_{\substack{Q_x \succeq 0 \\ \sum_{x \in G} Q_x = I}} \sum_{x \in G} \text{Tr}[Q_x \Gamma(S_r^\diamond(x))], \quad \diamond \in \{\exists, \forall\}. \quad (26)$$

This is a standard finite quantum reward program of the type due to Holevo and to Yuen–Kennedy–Lax.

Proposition 1 (Finite-report equivalence and compression). *Let $\mathcal{H}_{r,G}^\diamond = \{S_r^\diamond(x) : x \in G\}$. Then*

$$V_G^\diamond = V_\Gamma(\mathcal{H}_{r,G}^\diamond). \quad (27)$$

If G contains a representative of every realizable list, then $V_G^\exists = \bar{\eta}_C(r)$ and $V_G^\forall = \underline{\eta}_C(r)$.

Proof. Merge all Q_x whose reports induce the same list; this preserves feasibility and the objective. Conversely, assign each list effect to any one report in G that realizes the list. The two transformations are inverse at the level of objective value. $\square$

Thus the hypergraph is an exact compression of duplicate reward columns in a finite report program. Bavaresco et al. treat a broader continuous objective: a general Bayesian reward with joint optimization of the probe, ancilla, measurement, and report through

a quantum tester [19]. Their finite reduction approximates the parameter integral by point hypotheses and then either discretizes or optimizes the report space, producing a reward-matrix tester SDP. For discontinuous fixed-width confidence rewards, their convergence analysis controls an additive score error through the prior mass near the reward boundaries, with an explicit bound under regularity such as a bounded prior density.

For fixed-tolerance estimation, the present construction instead controls the approximation through the tolerance radius. It retains the exact prior-weighted cell integrals (5), converts possible and certain metric coverage into finite inner and outer programs, and compresses reports that induce the same reward column into one realized hyperedge. The resulting guarantee is horizontal rather than additive: the two programs sandwich the exact success probability after shifting the tolerance radius by at most Δ_C . This needs neither a prior density nor a modulus of continuity for the parameter-to-state map, although convergence is asserted only at continuity points and presupposes refinements with $\Delta_C \rightarrow 0$. Theorem 1 is stated for output states, while Corollary 1 applies the same geometry to the tester-level protocol optimization of Bavaresco et al. In this sense, the two frameworks are complementary rather than ordered.

3.1. Extension to quantum channel testers

Let $\mathcal{E}_\theta : \mathcal{L}(\mathcal{H}_I) \rightarrow \mathcal{L}(\mathcal{H}_O)$ be a channel with Choi operator C_θ . A one-use tester-valued measure $T(dx)$ satisfies $T(dx) \succeq 0$ and

$$\int_{\mathbf{X}} T(dx) = I_O \otimes \sigma, \quad \sigma \succeq 0, \quad \text{Tr } \sigma = 1, \quad (28)$$

under the usual Choi convention. Then $\text{Tr}[T(dx)C_\theta]$ is the report probability. Define

$$\Xi_k = \int_{C_k} C_\theta \mu(d\theta), \quad \Xi(S) = \sum_{k \in S} \Xi_k. \quad (29)$$

For a hypergraph $\mathcal{H}$, let

$$V_\Xi^{\text{test}}(\mathcal{H}) = \max_{\substack{T_S \succeq 0, \sigma \succeq 0 \\ \sum_S T_S = I_O \otimes \sigma, \text{Tr } \sigma = 1}} \sum_{S \in \mathcal{H}} \text{Tr}[T_S \Xi(S)]. \quad (30)$$

Corollary 1 (Extension to quantum testers). *Theorem 1 remains valid upon replacing POVMs and output states by tester-valued measures and Choi operators, respectively, and V_Γ by V_Ξ^{test} . All inner–outer, radius-shift, monotonicity, refinement, and convergence statements remain valid. The finite inner and outer programs are SDPs. The conclusion also holds for any higher-order tester class characterized by positivity and a common affine normalization.*

Proof. Every step of the proof of Theorem 1 uses only positivity, linearity of the score, aggregation or relabeling of decision effects, and a normalization shared by all reports. Tester elements have exactly these properties. Integrating Choi operators over each cell gives (29); grouping a continuous tester by a list preserves (28), and relabeling finite tester elements preserves the same affine normalization. The geometric inclusions are unchanged. Finite tester constraints are linear matrix constraints, hence the resulting programs are SDPs. $\square$

This corollary places the geometric certificate around the protocol optimization of [19]: the tester SDP there supplies the best score for a declared finite reward program, while the inner and outer programs certify how that score relates to the continuous fixed-tolerance task.

3.2. Comparison with list- M -ary reductions

Let

$$P_{\text{guess}}^* = \max_{\substack{N_k \succeq 0 \\ \sum_k N_k = I}} \sum_{k=1}^K \text{Tr}(\Gamma_k N_k) \quad (31)$$

be ordinary minimum-error discrimination success. For $\ell \geq 1$, let

$$\mathcal{L}_\ell = \{S \subseteq [K] : |S| \leq \ell\}. \quad (32)$$

Proposition 2 (Geometry versus fixed list size). *If $\ell(\mathcal{H}) = \max_{S \in \mathcal{H}} |S|$, then*

$$V_\Gamma(\mathcal{H}) \leq V_\Gamma(\mathcal{L}_\ell(\mathcal{H})) \leq \min\{1, \ell(\mathcal{H})P_{\text{guess}}^*\}. \quad (33)$$

Both inequalities can be strict.

Proof. The first inequality follows because $\mathcal{H} \subseteq \mathcal{L}_\ell(\mathcal{H})$. For the second, any effect assigned to the empty list can first be reassigned to an arbitrary nonempty list without decreasing the reward. We may therefore take $Q_\emptyset = 0$. Split each remaining list effect uniformly among its labels:

$$N_k = \sum_{S \ni k} \frac{Q_S}{|S|}. \quad (34)$$

Then $\{N_k\}$ is a singleton-decision POVM and

$$\begin{aligned} P_{\text{guess}}^* &\geq \sum_k \text{Tr}(\Gamma_k N_k) \\ &= \sum_{S \neq \emptyset} \frac{1}{|S|} \text{Tr}[Q_S \Gamma(S)] \geq \frac{1}{\ell} \sum_S \text{Tr}[Q_S \Gamma(S)]. \end{aligned} \quad (35)$$

Maximization gives the second inequality; $V_\Gamma \leq 1$ follows from $\Gamma(S) \preceq \sum_k \Gamma_k$.

For strictness, take three singleton target cells at $0, 3/2, 3$, radius $r = 1$, identical output states, and prior masses $(0.45, 0.10, 0.45)$. Metric balls realize $\{1, 2\}$ and $\{2, 3\}$ but not $\{1, 3\}$. Hence $V_\Gamma(\mathcal{H}) = 0.55$, whereas an unconstrained list of size two chooses $\{1, 3\}$ and obtains $V_\Gamma(\mathcal{L}_2) = 0.90 = 2P_{\text{guess}}^*$. $\square$

When all Γ_k commute, the quantum program reduces exactly to a classical set-valued Bayes rule. If $q_k(y)$ is the prior-weighted output density for cell k , then

$$V_{\text{cl}}(\mathcal{H}) = \int \max_{S \in \mathcal{H}} \sum_{k \in S} q_k(y) \nu(dy). \quad (36)$$

For $\mathcal{L}_\ell$, the integrand is the sum of the ℓ largest posterior cell masses. This gives the receiver-matched list- M -ary comparator used below.

3.3. Moment bounds and computation

Let $Z = d(\hat{X}, \beta(\Theta))$. Since every estimator satisfies $\Pr\{Z > r\} \geq 1 - \bar{\eta}_C(r)$, the layer-cake identity gives, for $p > 0$,

$$\inf_M \mathbb{E} Z^p \geq \int_0^\infty p r^{p-1} [1 - \bar{\eta}_C(r)] dr. \quad (37)$$

This has the outer architecture of a Ziv–Zakai bound, but a geometry-constrained list decision replaces a binary test. Because $\bar{\eta}_C(r)$ is already nondecreasing, no separate valley-filling operation is required.

The finite hypergraph can still be large: in the worst case it has 2^K edges. Geometry often makes it smaller. For interval cells on a line, membership changes only at interval endpoints, so evaluating the endpoints and the open intervals between them gives at most $4K + 1$

lists; the equal-cell experiment below has exactly $2K$. In higher-dimensional target spaces the number of metric-ball intersection patterns can be exponential. Practical implementations may therefore require arrangement algorithms, adaptive edge generation, or cutting-plane methods that add only dual constraints violated by the current solution.

For qubit composite operators, there is an exact low-dimensional reduction.

Corollary 2 (Qubit reduction). *Write*

$$\Gamma(S) = \frac{1}{2} [w_S I + \mathbf{v}_S \cdot \boldsymbol{\sigma}], \quad (38)$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ is the vector of Pauli matrices. Then,

$$V_\Gamma(\mathcal{H}) = \min_{\mathbf{z} \in \mathbb{R}^3} \max_{S \in \mathcal{H}} \{w_S + \|\mathbf{z} - \mathbf{v}_S\|_2\}. \quad (39)$$

If all $\mathbf{v}_S$ lie in a linear subspace L , the minimization may be restricted to L .

Proof. Write the dual variable as $Y = \frac{1}{2} [tI + \mathbf{z} \cdot \boldsymbol{\sigma}]$, so $\text{Tr } Y = t$. The constraint $Y \succeq \Gamma(S)$ is equivalent to $t - w_S \geq \|\mathbf{z} - \mathbf{v}_S\|_2$. Minimizing over t gives (39). Orthogonal projection of $\mathbf{z}$ onto L cannot increase its distance to any $\mathbf{v}_S \in L$. $\square$

4. STYLIZED MAGNETIC-NAVIGATION EXAMPLE

4.1. Map, quantum sensor, and matched baselines

We use a stylized one-dimensional model of MagNav to illustrate the effect of map aliases. Position $x \in [0, 4]$ is measured in map periods, with prior

$$p(x) = \frac{1}{2} \sin^2\left(\frac{\pi x}{4}\right), \quad (40)$$

and magnetic map

$$m_\epsilon(x) = \cos(2\pi x) + \epsilon x. \quad (41)$$

At $\epsilon = 0$, four exact map-period aliases lie in the prior support. A weak linear trend progressively breaks the global symmetry without changing the local sensitivity much.

At interrogation t , the system is in the pure Ramsey state

$$|\psi_t(x)\rangle = \frac{|0\rangle + e^{i\phi_t(x)}|1\rangle}{\sqrt{2}}, \quad \phi_t(x) = 1.5m_\epsilon(x + 0.173t). \quad (42)$$

For the multi-interrogation experiment we fix the four-outcome equatorial POVM

$$E_j = \frac{1}{4} [I + \cos(\pi j/2)\sigma_x + \sin(\pi j/2)\sigma_y], \quad (43)$$

independently at each time. Its phase Fisher information is one for every phase, equal to the pure-state phase QFI, using the continuous limiting value at zero-probability outcomes. Thus the local comparison is not weakened by a deliberately inefficient receiver.

The prior Fisher information and output QFI are

$$\begin{aligned} J_p &= \int_0^4 p(x) [\partial_x \log p(x)]^2 dx = \frac{\pi^2}{4}, \\ J_Q(x) &= 1.5^2 \sum_{t=0}^{T-1} [m'_\epsilon(x + 0.173t)]^2. \end{aligned} \quad (44)$$

The scalar van Trees relaxation used below is therefore

$$\text{MSE} \geq \left[J_p + \int_0^4 p(x) J_Q(x) dx \right]^{-1}. \quad (45)$$

The chosen POVM has the same local Fisher information, so (45) is also the local Bayesian Cramér–Rao bound for the fixed receiver, subject to the regularity conditions [20, 21].

To keep global comparisons operationally matched, let $q_T(y|x)$ be the complete outcome law of (43). The receiver-specific binary Ziv–Zakai testing function is

$$f_M(\tau) = \frac{1}{2} \int_0^{4-\tau} [p(x) + p(x+\tau) - \|p(x)q_T(\cdot|x) - p(x+\tau)q_T(\cdot|x+\tau)\|_1] dx. \quad (46)$$

With valley filling $\mathcal{V}f(\tau) = \sup_{s \geq \tau} f(s)$,

$$\Pr\{|\hat{x} - x| \geq r\} \geq \mathcal{V}f_M(2r), \quad \text{MSE} \geq \frac{1}{2} \int_0^\infty \tau \mathcal{V}f_M(\tau) d\tau. \quad (47)$$

The measurement-independent quantum comparator replaces the weighted output laws in (46) by the corresponding weighted product states and the classical L_1 distance by trace norm. The fixed-cardinality list comparator uses (36) with all subsets of size at most $L(r)$, the largest outer metric-list size at radius r .

4.2. Receiver-matched results

Figure 1 compares only quantities for the fixed POVM, except that panel (b) also displays the weaker measurement-independent quantum binary QZZB for context. In panel (a), all tail bounds are receiver matched. At $\epsilon = 0$ and $T = 6$, the numerical Bayes failure is 0.55575. The matched binary ZZ lower bound is 0.52504, the unconstrained list-18 lower bound is 0.30072, and the geometry-resolved interval is

$$0.54983 \leq P_{\text{fail},M}^*(0.2) \leq 0.55629. \quad (48)$$

At $\epsilon = 0.6$, the corresponding values are 0.36788, 0.24003, 0.20137, and

$$0.36035 \leq P_{\text{fail},M}^*(0.2) \leq 0.36855. \quad (49)$$

The full hypergraph remains informative because it retains which groups of cells can be covered by one interval. The unconstrained list comparator may select high-posterior cells from several mutually incompatible map periods.

The MSE comparison in Table 1 emphasizes the local/global distinction. At $T = 6$, the local bound is 3.72×10^{-3} , whereas the numerical Bayes MSE is 0.52261, about 140 times larger. The quantum binary QZZB raises the lower bound to 0.32281; imposing the stated receiver raises the matched binary ZZB to 0.33755; retaining the full metric hypergraph raises it further to 0.34505. The fixed-cardinality list moment bound becomes weak because it may assemble a list from separated aliases.

Table 2 separates the operational classes for fixed-radius failure. The near equality of the quantum and matched binary values at $\epsilon = 0$ shows that receiver mismatch was not driving the original binary/hypergraph gap in the exactly periodic case. The gap persists because binary testing does not retain the complete multiway list geometry. As T grows, the unconstrained list- M -ary bound becomes especially weak, while the metric hypergraph stays close to the numerical Bayes optimum.

4.3. Unrestricted quantum converse and radius dependence

Figure 2(a) keeps the receiver fixed and compares the entire failure curves for one interrogation. Panel (b) instead removes the receiver restriction for the converse. Corollary 2 evaluates the one-interrogation

quantum hypergraph program as a two-dimensional weighted covering problem. At $R = 0.2$,

$$0.62591 \leq P_{\text{fail},Q}^*(0.2) \leq 0.65609, \quad (50)$$

whereas the binary quantum ZZ tail lower bound is 0.52724. Hence the geometry-resolved improvement is not merely a consequence of fixing a classical receiver. Integrating the quantum lower curve using (37) gives an MSE lower bound 0.42331, compared with 0.36681 for the binary QZZB. For the fixed POVM, the matched binary, unconstrained-list, hypergraph, and achieved MSE values are respectively 0.42043, 0.31973, 0.45353, and 0.52276.

The outer hypergraph program is also exactly the direct finite-report reward program of Proposition 1 after duplicate columns are merged. For $K = 160$ and $R = 0.2$, the continuous interval arrangement has 320 realized outer lists and maximum list size 18. This replaces treating each of the 1601 quadrature locations as a separate report label by one decision effect per distinct geometric reward column. Table 3 verifies monotone tightening and shows the linear list count for this one-dimensional geometry.

4.4. Interpretation and limitations

The stylized MagNav example shows something that continues to hold in higher-dimensional systems: local field sensitivity and global map identifiability are different resources. In more realistic settings, θ may include the complete path, attitude, inertial errors, map realization, sensor bias, altitude, and platform interference, while $\beta(\theta)$ may be terminal position or a full path equipped with a maximum-deviation metric. Nuisance uncertainty is averaged inside Γ_k or Ξ_k before distinguishability is assessed. No additive trajectory displacement, trajectory density, or differentiable observation law is required.

The computational difficulty can nevertheless grow sharply. Interval cells give a linear number of realized lists and commuting receiver outputs reduce the decision to posterior mass calculations. General two- or three-dimensional trajectory clusters may generate many more metric-ball incidence patterns, and unrestricted multi-interrogation quantum outputs enlarge the matrix dimension. A practical higher-dimensional implementation would likely combine particle or trajectory clustering with adaptive hyperedge generation, rather than enumerate 2^K subsets. Establishing such scalable algorithms and applying the tester version to realistic vector or gradiometric magnetometers are natural next steps.

5. CONCLUSION

The useful estimation question in an aliased map is not only how sharply one can refine the correct local mode, but how much posterior mass can be placed inside one admissible protection ball. Established Bayesian quantum decision theory gives the continuous reward problem and the finite tester SDP once reports and parameters are discretized. The contribution here is a certified geometric bridge between them: exact cell composites, inner and outer metric-list programs, a target-diameter radius guarantee, monotone refinement, and a direct tester lift. In the stylized MagNav example, this bridge separates local sensitivity from wrong-region lock. The QFI/Bayesian Cramér–Rao bound becomes extremely small while global Bayes error remains large. Matched binary and unconstrained list bounds recover only part of the missing information, whereas the full metric hypergraph nearly certifies the fixed-radius risk. The same construction applies whenever several latent hypotheses may be jointly acceptable under a task metric and the geometry of those acceptable sets matters to the decision.

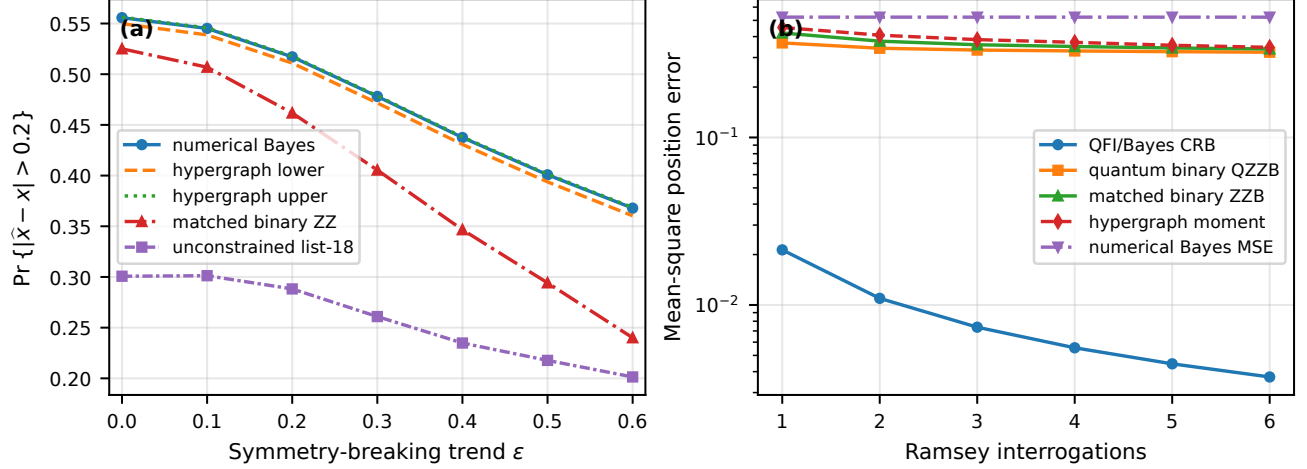


Fig. 1. Receiver-matched MagNav comparison. (a) Six interrogations and $R = 0.2$: as the trend breaks periodic aliases, the metric-hypergraph bracket follows numerical Bayes failure. The binary ZZ curve is computed from the same fixed-POVM output law, while the unconstrained list-18 curve retains list cardinality but discards metric realizability. (b) Exactly periodic map: the local QFI/Bayesian Cramér–Rao bound collapses with interrogation count, although the Bayes MSE remains near 0.523. The fixed-receiver hypergraph moment bound is stronger than the matched binary ZZB; the measurement-independent quantum QZZB is also shown.

Table 1. MSE lower bounds and numerical Bayes MSE for the exactly periodic map, $\epsilon = 0$. “Q-bin” is the measurement-independent quantum binary QZZB; “M-bin” is the binary ZZB matched to the fixed POVM; “list” is the unconstrained fixed-cardinality list bound; and “HG” is the fixed-POVM hypergraph moment bound.

T	QFI/BCRB	Q-bin QZZB	M-bin ZZB	list moment	HG moment	numerical Bayes MSE
1	0.02133	0.36681	0.42043	0.31973	0.45353	0.52276
2	0.01095	0.34085	0.37574	0.21332	0.40796	0.52270
3	0.00737	0.33247	0.35760	0.14800	0.38373	0.52266
4	0.00555	0.32844	0.34919	0.10981	0.36966	0.52266
5	0.00445	0.32514	0.34213	0.08222	0.35505	0.52263
6	0.00372	0.32281	0.33755	0.06335	0.34505	0.52261

6. ACKNOWLEDGMENTS

ChatGPT 5.6-Sol was used to support experiment implementation, mathematical development, and writing.

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Table 2. Failure-probability bounds at $R = 0.2$ for the exactly periodic map. “Q-bin” is measurement independent; all other bounds and numerical Bayes values use the fixed POVM.

T	Q-bin lower	M-bin lower	list- L lower	HG lower	HG upper	numerical Bayes
1	0.52724	0.53329	0.65853	0.69836	0.72061	0.71956
2	0.52558	0.52820	0.55186	0.63680	0.65373	0.65243
3	0.52536	0.52724	0.46758	0.59914	0.61243	0.61151
4	0.52524	0.52677	0.40459	0.57817	0.58888	0.58805
5	0.52501	0.52578	0.34762	0.56111	0.56942	0.56876
6	0.52483	0.52504	0.30072	0.54983	0.55629	0.55575

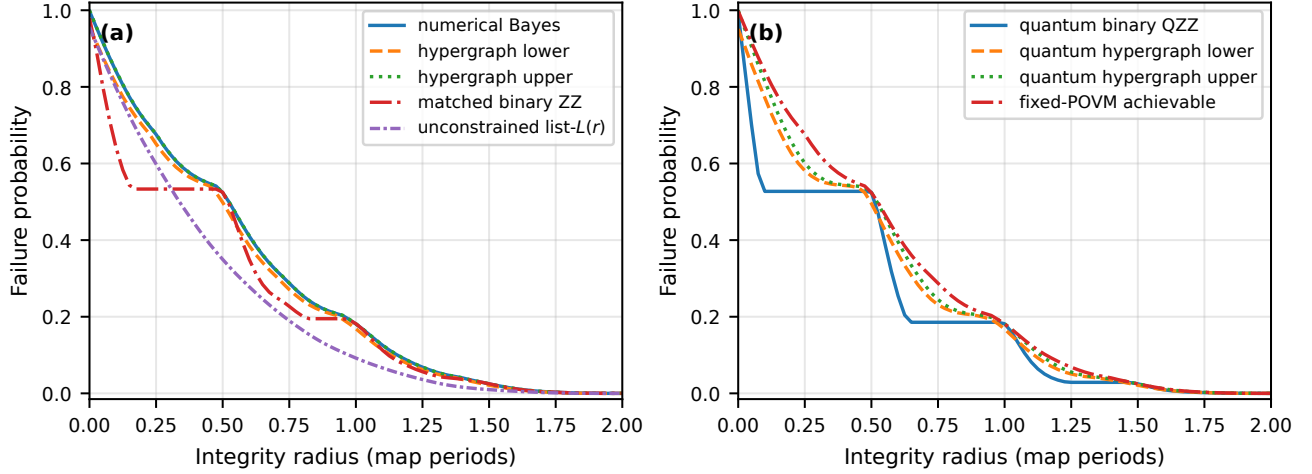


Fig. 2. One-interrogation radius dependence at $\epsilon = 0$. (a) Fixed receiver: numerical Bayes failure, geometry-resolved bracket, receiver-matched binary ZZ, and unconstrained list- $L(r)$ bounds. (b) Measurement-independent quantum converse: the quantum hypergraph bracket sharpens the binary QZZ tail bound; fixed-POVM Bayes performance is an achievable reference, not a bound on the same measurement class. Plateaus occur at radii set by the repeated map periods.

Table 3. Cell refinement for $T = 6$, $\epsilon = 0$, and $R = 0.2$. “Bayes” is the numerical fixed-POVM value.

K	Δ_C	edges	$L_{\max}$	HG LB	Bayes	HG UB
40	.1000	80	6	.53544	.55575	.55891
80	.0500	160	10	.54459	.55575	.55687
160	.0250	320	18	.54983	.55575	.55629
320	.0125	640	34	.55265	.55575	.55612

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